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Sensitivity of Singularity Points Position to Fracture Weakness Change

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Introduction

Fractures play a crucial role in controlling subsurface fluid flow. Their presence enhances permeability and storage capacity, making them critical targets in reservoir characterization. However, when their dimensions are smaller than the seismic wavelength, direct detection becomes problematic. This limitation imposes critical challenges on seismic interpretation and reservoir characterization [1, 2]. As shown by [3], such sub-wavelength fractures often elude conventional seismic methods. Effective medium theories, such as the Vertical Fracture Transverse Isotropy (VFTI Medium), which assumes vertically aligned fractures and a vertical transverse isotropy to represent seismic anisotropy in fractured media. [4, 5, 6, 10]. As a subset of orthorhombic (ORT) symmetry, VFTI includes fracture weakness parameters that quantify directional stiffness reductions [5]. The ORT model predicts the singularity point where S1 and S2 shear wave velocities coincide [7]. This study investigates how these singularity points shift with variations in fracture weakness.

Method and Theory

Schoenberg and Helbig [5] showed that vertical fractures combined with fine horizontal layering form a long-wavelength orthorhombic medium, equivalent to fractures embedded in a VTI background. Their model remains a standard framework for describing seismic anisotropy and interpreting fractured reservoirs.

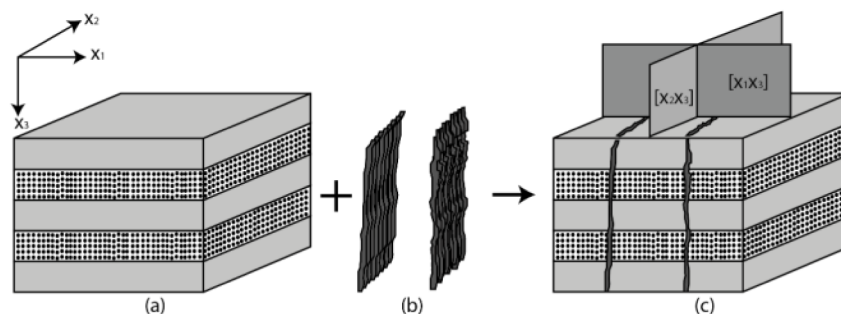


Figure 1—Schematic diagram of vertical fractures embedded in VTI medium adopted from [5]. (a) VTI medium, (b) Vertical fractures set, and (c) ORT medium. [5]

Here, an agreement to the symmetry planes is set, where $[x_1, x_2]$ and $[x_2, x_3]$ are the vertical symmetry planes that are perpendicular to the fractures and parallel to the fractures, respectively. As for the $[x_1, x_3]$ is the horizontal symmetry plane. A transversely isotropic background medium is assumed, typically with a vertical symmetry axis (VTI), which is characterised by five independent elastic constants. When a set of aligned, vertical fractures is introduced into this VTI background, the effective medium symmetry reduces to orthorhombic (ORT). This occurs because the fractures break the azimuthal isotropy of the horizontal plane, while the VTI background already distinguishes vertical from horizontal properties. Importantly, this is not a generic ORT medium that can be defined arbitrarily, but rather one constrained by its origin as the combination of a VTI background and a single system of vertical fractures.

The concept of fracture weaknesses, defined as dimensionless measures of stiffness reduction caused by the presence of fractures, was introduced in [5]. Two main types are distinguished: the normal weakness (δ_N), which controls the compliance effect in the direction normal to the fracture plane, and the shear weakness (δ_T), which controls the compliance effect in the tangential or sliding direction along the fracture. These parameters quantify the extent to which fractures perturb the background stiffnesses. They are always non-negative and typically small ($\delta \leq 1$), ensuring that fractures act as perturbations rather than dominating the host medium. These values represent a weakly fractured medium, where fractures do not overwhelm the stiffness of the background but are sufficiently influential to induce measurable anisotropy. By adopting such values, the authors were able to analytically explore the behaviour of phase velocities, slowness surfaces, and singularity points within the effective orthorhombic model.

Referring to the Schoenberg-Helbig model, a VTI background medium with one vertical fracture set is considered. The fracture system is embedded by introducing three fracture weakness elements. Here, δ_N , δ_V , and δ_H represent the strain contributions that result from the normal, vertical tangential, and horizontal tangential fracture compliances, respectively. In an orthorhombic medium, there is always at least one singularity located within one of the symmetry planes. The key challenge lies in determining not only which plane this singularity resides in, but also the conditions necessary to identify and select the appropriate symmetry plane, specifically in the context of a Vertically Fractured Transversely Isotropic (VFTI) medium. The generic form gives the stiffness coefficient matrix C for the ORT model:

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & & & \\ & c_{22} & c_{23} & & & \\ & & c_{33} & & & \\ & & & c_{44} & & \\ & & & & c_{55} & \\ & & & & & c_{66} \end{bmatrix} \quad (1)$$

Here, c_{ij} are the independent stiffness coefficients in the two-index Voigt notation. This discussion begins with the investigation of wave phenomena in anisotropic media, focusing on the Christoffel equation. This matrix is associated with the eigenvalues corresponding to the phase velocity of the three different wave modes. In fracturing media, and especially in orthogonally symmetric media, these singularity points tend to be off-axis, mostly due to the anisotropy introduced by fractures. At certain values, two eigenvalues become equal, and their corresponding eigenvectors become dependent, signaling a degeneracy of the wave modes. To determine the slowness surfaces, we describe singularity points as regions where the slowness gradient of two wave modes becomes zero, causing these surfaces to converge or touch.

Following the Schoenberg-Helbig model [5], wherein the background medium is characterized as a Vertical Transversely Isotropic (VTI) medium, a single vertical fracture set is introduced. This incorporation involves three fracture weakness elements: δ_N , representing the strain resulting from normal fracture compliance; δ_V , the strain arising from vertical tangential fracture compliance; and δ_H .

The formation of an equivalent Orthorhombic (ORT) medium, subject to the long-wavelength limit, is attributed to the combined influence of both horizontal fine layering and vertical fractures, as elucidated

[5]. Following the ORT model, wherein the background medium is characterized as a Vertical Transversely Isotropic (VTI) medium, a single vertical fracture set is introduced. This incorporation involves three fracture weakness elements: δ_N , representing the strain resulting from normal fracture compliance; δ_V , the strain arising from vertical tangential fracture compliance; and δ_H , denoting the strain originating from horizontal tangential fracture compliance. Schoenberg and Sayers derived an expression elucidating the equivalent medium properties by augmenting the fracture compliance matrix with the background compliance matrix, as detailed [5]. The resulting expression for the stiffness matrix of the effective medium is provided, and the individual components of the stiffness matrix can be written in the c_{ij} , $i, j = 1, \dots, 6$.

$$\begin{aligned}
 c_{11} &= c_{11b}(1 - \delta_N), \\
 c_{12} &= (c_{11b} - 2c_{66b})(1 - \delta_N), \\
 c_{13} &= c_{13b}(1 - \delta_N), \\
 c_{22} &= c_{11b} \left(1 - \delta_N \frac{(c_{11b} - 2c_{66b})^2}{c_{11b}^2} \right), \\
 c_{23} &= c_{13b} \left(1 - \delta_N \frac{(c_{11b} - 2c_{66b})}{c_{11b}} \right), \\
 c_{33} &= c_{33b} \left(1 - \delta_N \frac{c_{13b}^2}{c_{11b}^2} \right), \\
 c_{44} &= c_{44b}, \\
 c_{55} &= c_{44b}(1 - \delta_V), \\
 c_{66} &= c_{66b}(1 - \delta_H).
 \end{aligned}$$

Following the approach [4], the equivalent medium is obtained by adding the fracture compliance matrix to the background compliance matrix, recollecting all the component the stiffness coefficient matrix can be represented as [equation 2](#). The corresponding stiffness matrix of the effective medium is expressed as

$$C = \begin{bmatrix} C_{11b}(1 - \delta_N) & C_{12b}(1 - \delta_N) & C_{13b}(1 - \delta_N) & 0 & 0 & 0 \\ 0 & C_{11b} \left(1 - \delta_N \frac{C_{12b}^2}{C_{11b}^2} \right) & C_{13b} \left(1 - \delta_N \frac{C_{12b}}{C_{11b}} \right) & 0 & 0 & 0 \\ 0 & 0 & C_{33b} \left(1 - \delta_N \frac{C_{13b}^2}{C_{11b} C_{33b}} \right) & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44b} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44b}(1 - \delta_V) & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66b}(1 - \delta_H) \end{bmatrix}. \quad (2)$$

Here, C_{ijb} are the stiffness coefficients of the VTI background explained in [Equation 2](#). δ_N , δ_V , and δ_H are the dimensionless weaknesses of the fractures, which range from zero (no fracture) to unity (extreme fracturing). δ_V and δ_H correspond to vertical and horizontal tangential fracture weaknesses, respectively, and δ_N corresponds to the normal fracture weakness. The tangential weaknesses δ_V and δ_H provide a measure of crack density, whereas the normal weakness δ_N provides information regarding the fluid content of the fractures and the possibility of fluid flow between the fractures and pore space[?]. Note this [equation 2](#) describes a particular type of orthorhombic medium, with stiffness satisfying the relation (Schoenberg and Helbig, 1997):

$$C_{13}(C_{22} + C_{12}) = C_{23}(C_{11} + C_{12}) \quad (3)$$

The existence of this additional constraint arises from the fact that, while a general orthorhombic medium is described by nine independent values of C_{ij} , the vertical fracture-only eight quantities define induced model considered here. These include five stiffness coefficients of the VTI background medium and the three fracture weaknesses (δN , δV , and δH).

In an orthorhombic medium, there is always at least one singularity located within one of the symmetry planes. The key challenge lies in determining not only which plane this singularity resides in, but also the conditions necessary to identify and select the appropriate symmetry plane, specifically in the context of a Vertically Fractured Transversely Isotropic (VFTI) medium.

The Christoffel matrix is given by $\mathbf{G} = G_{ij} = c_{ijkl}p_j p_k$, where c_{ijkl} is the stiffness coefficient tensor, and $p_{ij}, i, j = 1, 2, 3$, are the projections of the slowness vector $\mathbf{p} = (p_1, p_2, p_3)^T$.

$$\begin{aligned} G_{11} &= c_{11}p_1^2 + c_{66}p_2^2 + c_{55}p_3^2, \\ G_{22} &= c_{66}p_1^2 + c_{22}p_2^2 + c_{44}p_3^2, \\ G_{33} &= c_{55}p_1^2 + c_{44}p_2^2 + c_{33}p_3^2, \\ G_{12} &= (c_{12} + c_{66})p_1 p_2 = d_{12}p_1 p_2, \\ G_{13} &= (c_{13} + c_{55})p_1 p_3 = d_{13}p_1 p_3, \\ G_{23} &= (c_{23} + c_{44})p_2 p_3 = d_{23}p_2 p_3. \end{aligned}$$

The characteristic (Christoffel) equation is by, given by

$$F = F(\mathbf{p}) = \det(\mathbf{G} - \mathbf{I}) = 0 \quad (4)$$

where $\mathbf{I}$ is the unit 3×3 matrix.

Singularity points are locations within a medium where seismic wave behaviour becomes highly sensitive to fracture parameters. These points are characterized by the coincidence of shear wave velocities, often resulting in non-differentiable wavefronts and unique seismic signatures [3]. Variations in fracture parameters can significantly shift the position of these singularities, thereby modifying wave velocities and seismic responses across different propagation directions [5, 8]. To investigate the sensitivity, we adopt the standard orthorhombic (ORT) model described by Schoenberg and Helbig [5]. In this model, the coordinate axes (1, 2, and 3) represent the principal directions of symmetry in the medium, giving rise to three primary symmetry planes: 13, 12, and 23. In our analysis, we examine these symmetry planes and the planes between them. Figure 2 presents three scenarios where fracture weaknesses are varied individually.

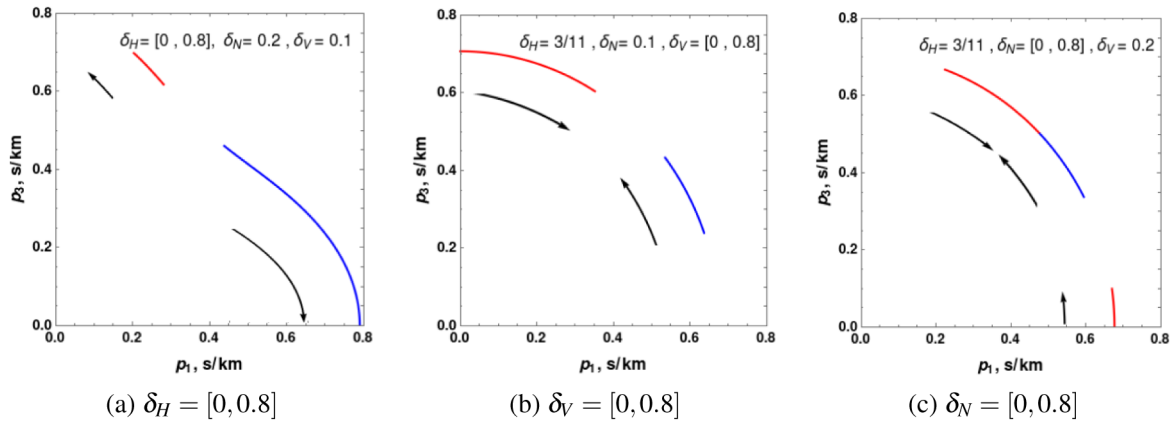


Figure 2—Symmetry Plane 13, positions of two singularity points [red (P3) and blue (P1)]. The arrows indicate the direction of movement of the singularity point as the corresponding fracture weakness increases. a) We fix two fracture weaknesses ($\delta_N = 0.1$, $\delta_V = 0.2$) and use $\delta_H = (0, 0.8)$ as the free parameter, b) ($\delta_N = 0.1$, $\delta_H = 0.273$) and use $\delta_V = (0, 0.8)$ as the free parameter, c) ($\delta_H = 0.273$, $\delta_V = 0.2$) and use $\delta_N = (0, 0.8)$ as the free parameter. Increasing δ_H causes the singularity points to migrate away from the origin along curved paths, indicating a growing anisotropic response in the horizontal direction. δ_N produces a noticeable but differently shaped trajectory as well. The changes in δ_V cause a symmetric shift of singularity points along the vertical axis, which leads to distinct trajectories of singularity points within the symmetry planes.

ORT media that have two or three values of the d_{12} , d_{13} , d_{23} , equal to zero, are degenerate with more complicated singular curves than in the case described above. If one of the d_{12} , d_{13} , d_{23} is equal to zero, the ORT model contains only discrete singular points located in the planes of symmetry. The singularity point (or singular direction) is defined by the slowness vector $\mathbf{ps} = (p_{1s}, p_{2s}, p_{3s})^T$, which for the Christoffel matrix defined by equation (4) gives the multiple roots. For this point, the phase velocities of S1 and S2 waves coincide. Theoretically, we can have a singularity point for S1 and P waves. If we have multiple eigenvalues, all second-order minors of the matrix $\mathbf{G} - \mathbf{I}$ are zero. Alternatively, at a singularity point, the characteristic equation $F(p_1, p_2, p_3) = 0$ has a multiple root concerning any of the variables p_1, p_2, p_3 .

$$F(p_1, p_2, p_3) = (-1 + c_{11}p_1^2 + c_{66}p_2^2 + c_{55}p_3^2)(-1 + c_{66}p_1^2 + c_{22}p_2^2 + c_{44}p_3^2)(-1 + c_{55}p_1^2 + c_{44}p_2^2 + c_{33}p_3^2) \\ + d_{12}^2 p_1^2 p_2^2 (-1 + c_{55}p_1^2 + c_{44}p_2^2 + c_{33}p_3^2) + d_{13}^2 p_1^2 p_3^2 (-1 + c_{66}p_1^2 + c_{22}p_2^2 + c_{44}p_3^2) \\ + d_{23}^2 p_2^2 p_3^2 (-1 + c_{11}p_1^2 + c_{66}p_2^2 + c_{55}p_3^2) - 2d_{12}d_{13}d_{23}p_1^2 p_2^2 p_3^2 = 0 \quad (5)$$

To identify the singularity points in an orthorhombic medium, we follow [8]. In a symmetry plane, the characteristic equation (10) is factorized. For example, in the 1-3 symmetry plane ($p_2 = 0$), it takes the form:

$$F(p_2 = 0) = (-1 + c_{66}p_1^2 + c_{44}p_3^2)(d_{13}^2 p_1^2 p_3^2 - (-1 + c_{55}p_1^2 + c_{33}p_3^2)(-1 + c_{11}p_1^2 + c_{55}p_3^2)) = 0. \quad (6)$$

Solving this as a system of equations, with the first equation $-1 + c_{66}p_1^2 + c_{44}p_3^2 = 0$ and the second equation $d_{13}^2 p_1^2 p_3^2 - (-1 + c_{55}p_1^2 + c_{33}p_3^2)(-1 + c_{11}p_1^2 + c_{55}p_3^2) = 0$, gives the slowness projections p_1 and p_3 for singularity points (0, 1, 2) in the 1-3 symmetry plane.

Similar solutions can be obtained for the remaining symmetry planes, namely 2-3 and 1-2. For a singularity point located between the symmetry planes, one must solve the system of equations $S_{jk} = 0$, $j, k = 1, 2, 3, j \neq k$, where S_{jk} denotes the jk off-diagonal minor of the Christoffel matrix (Equation 2). This system needs to be solved for all slowness projections p_1, p_2 , and p_3 .

To find the group velocity image of the singularity point, I begin with the characteristic polynomial, which can be expanded into a Taylor series in the vicinity of a given point defined by the vector $\mathbf{ps} = (p_{1s}, p_{2s}, p_{3s})^T$.

In-between symmetry planes

The Christoffel tensor $\mathbf{G}$ and the characteristic equation can also be rewritten in terms of the phase velocity v by substituting $p_j = n_j/v$, where n_j are the projections of the directional vector in the phase domain. The singularity points can be found by setting all the second-order minors of the Christoffel tensor to zero, $M_{jk} = 0$. For orthorhombic (ORT) media, we have one essential (fundamental) symmetry plane and two non-essential symmetry planes.

In the first quadrant of the essential symmetry plane, there is always one singularity point regardless of the value of an elliptic parameter defined in this plane. The other (non-essential) symmetry planes can have zero, one, or two singularity points depending on the value of the corresponding anelliptic parameter. Additionally, a singularity point may be located in-between the symmetry planes.

To find the group velocity image of the singularity point, I begin with the characteristic polynomial, which can be expanded into a Taylor series in the vicinity of a given point defined by the vector $\mathbf{p}_s = (p_{1s}, p_{2s}, p_{3s})^T$ as the symmetry planes of ORT, $d_{jk} > 0$, nondiagonal minors:

$$\begin{aligned} \left(c_{11} - \frac{d_{12}d_{13}}{d_{23}}\right)p_1^2 + c_{66}p_2^2 + c_{55}p_3^2 - 1 &= 0; \\ c_{66}p_1^2 + \left(c_{22} - \frac{d_{12}d_{23}}{d_{13}}\right)p_2^2 + c_{44}p_3^2 - 1 &= 0; \\ c_{55}p_1^2 + c_{44}p_2^2 + \left(c_{33} - \frac{d_{13}d_{23}}{d_{12}}\right)p_3^2 - 1 &= 0; \end{aligned} \quad (7)$$

Slowness projection:

$$p_1^2 = \frac{D_1}{D}, \quad p_2^2 = \frac{D_2}{D}, \quad p_3^2 = \frac{D_3}{D}$$

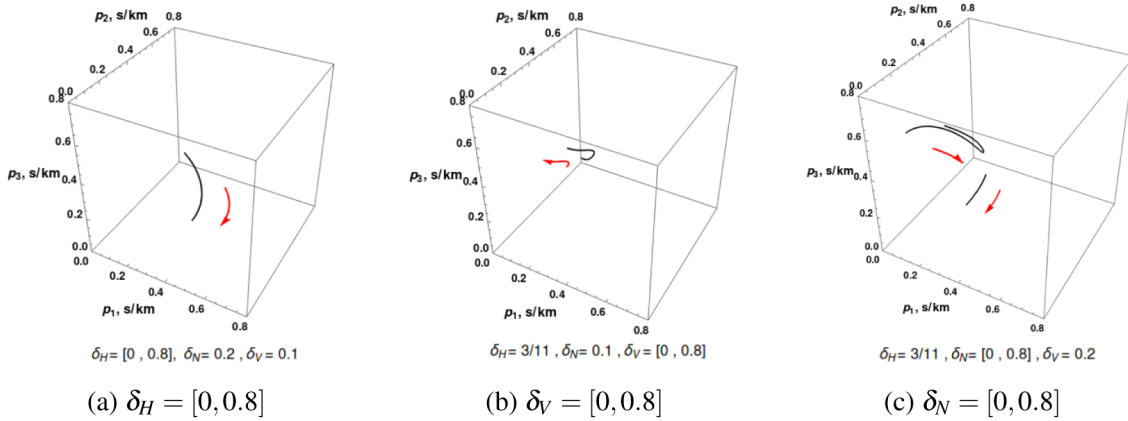


Figure 3—Position of two singularity points (shown by red and blue colours) in-between the symmetry planes with the change in fracture weaknesses δ_H (left), δ_V (middle), and δ_N (right). The arrows indicate the direction of the singularity point movement when the corresponding fracture weakness increases.

A singularity point with conical degeneracy gives rise to an ellipse [8], corresponds to how fast a seismic wave travels through the medium, group velocity image of the singularity point is given by an ellipse. In group velocity ellipses are practically insensitive to changes in δ_N in Figure 4. However, δ_V and δ_H , the ellipses become progressively more elongated in opposite directions. A singularity point with conical degeneracy gives rise to an ellipse in the group velocity domain [8].

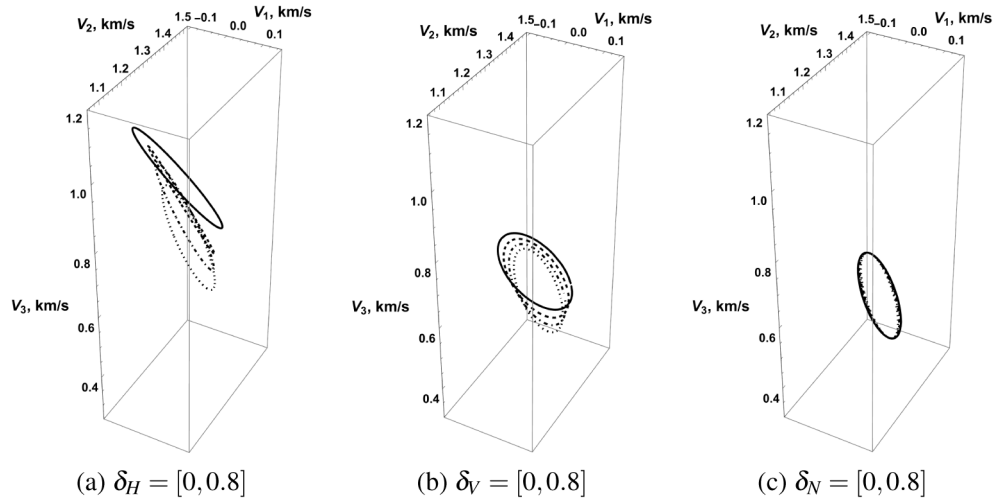


Figure 4—Group velocity ellipses in response to fracture weaknesses in symmetry Plane 12 : (a) δ_H induces a smooth and progressive change in ellipse shape; (b) δ_V produces only minor axis adjustments along the axes; (c) δ_N causes slight, often negligible, variations.

The group velocity ellipses corresponding to singularity points in symmetry plane 13 (perpendicular to the fracture planes), computed for different values of fracture weaknesses, are shown in Figure 4. It can be observed that the group velocity ellipses are practically insensitive to changes in δ_N . However, as the other fracture weaknesses increase, the ellipses become progressively more elongated in opposite directions.

The group velocity ellipses for a singularity point located between the symmetry planes are shown in Figure 5. A significant change in the shape of the ellipses is observed with increasing δ_H and δ_V , while they remain largely unaffected by variations in δ_N .

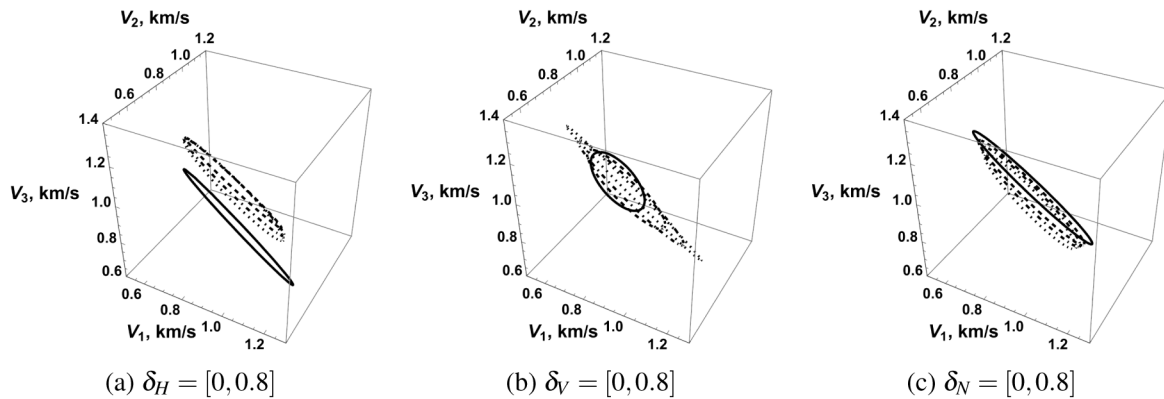


Figure 5—Response of group velocity ellipses to fracture weaknesses for a singularity point located between symmetry planes: (a) increasing δ_H produces a level change of the ellipse and smooth change in ellipse shape, (b) increasing δ_V causes elongated ellipse along the axis, and (c) variations in δ_N lead to slight, generally negligible, changes.

The size of the anomaly in the group velocity domain and offset domain can be computed as the square of the corresponding ellipses, as described by Stovas [9].

$$S = \pi \sqrt{\frac{\mathbf{P}_s^2 \det \mathbf{M}}{(\mathbf{P}_s^T \mathbf{M} \mathbf{P}_s)^3}}, \quad S_{XY} = \pi Z^2 \sqrt{\frac{\det \mathbf{M}}{m_{33}^3}}$$

Here, $\mathbf{P}_s = (P_1^S, P_2^S, P_3^S)^T$ is the slowness vector evaluated at the singularity point, and $\mathbf{M} = (m_{ij})$ is the 3×3 matrix composed of the second-order partial derivatives of the Christoffel polynomial to the slowness components, also evaluated at the singularity point. The variable $Z = 1 \text{ km}$ the depth coordinate.

The term S represents the spatial extent of the anomaly in the group velocity domain (measured in km^2/s^2), while S_{XY} represents its projected size in the horizontal (offset) domain (measured in km^2). These expressions provide a geometric measure of how strongly the singularity affects wave propagation characteristics. Figure 6 illustrates the dependence of S and S_{XY} on fracture weaknesses for a singularity point located between symmetry planes. From this analysis, it is evident that the anomaly associated with the singularity can be substantial. In particular, the projected area S_{XY} can reach values as high as 0.25-0.3 km^2 , which falls within the detection limits of modern seismic imaging techniques.

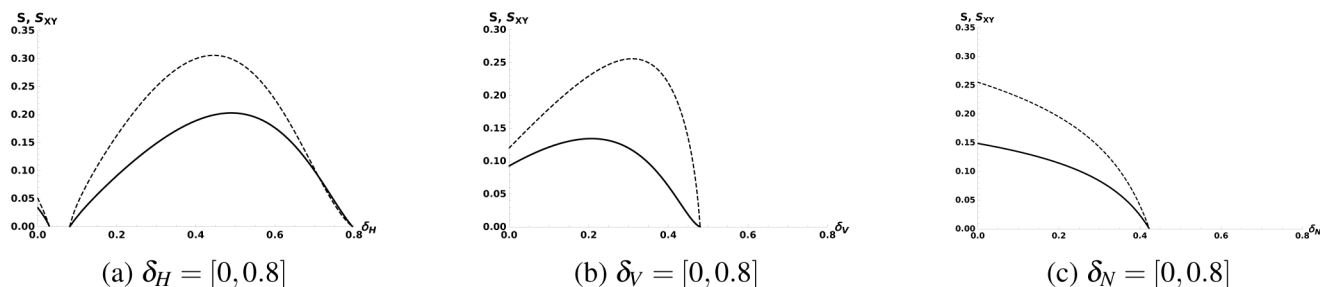


Figure 6—Changes in S and S_{XY} computed for a singularity point located between the symmetry planes, shown as functions of fracture weaknesses: δ_H (left), δ_V (middle), and δ_N (right). The anomaly size in the group velocity domain, S (km^2/s^2), is represented by bold lines, while the projected size in the offset domain, S_{XY} (km^2), is shown with dashed lines.

The geometry of the ellipse, figure 7, illustrates the anisotropic nature of the system's sensitivity to variations in fracture properties. The elongated ellipse suggests that the singularities exhibit heightened sensitivity to perturbations along particular orientations of fracture weakness; the ellipse's major axis defines the principal direction of sensitivity, signifying the orientation along which even minor changes in fracture characteristics induce the most substantial displacements in the position of the singularities.

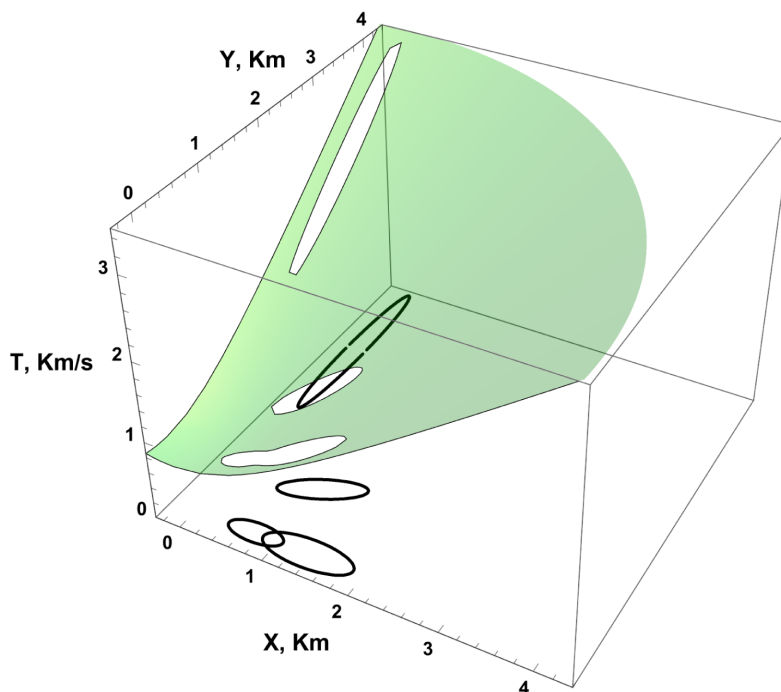


Figure 7—Illustration of the displacement of singularity points in response to variations in fracture characteristics. The surrounding ellipse represents the sensitivity to changes in all directions, with its shape and orientation indicating the degree and directionality of anisotropy in the system's response.

Conclusions

This study demonstrates the existing singularity points in 1-3, 2-3, and in-between symmetry plane, anomaly linked to singularity points is practically insensitive to changes in fracture weakness, denoted as δ_N , and more sensitive to δ_H . This analysis reveals that these anomalies are detectable in seismic data and can serve as additional constraints for the inversion of fracture parameters.

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